



# STROKE PATIENT TRANSPORT PROTOCOL

Document ID: SU-002-PROT-03 | Parent Policy: SU-002 — Stroke Code Activation — Pre-Hospital and In-Hospital | Version: 1.0 | Controlled Copy

Prehospital and Interfacility Transport — Saudi Red Crescent Authority (SRCA) & Stroke System of Care

## Document Control

- Document Number: SU-002-PROT-03
- Title: Stroke Patient Transport Protocol
- Effective Date: [To be assigned upon approval]
- Review Date: Annually
- Approved By: Stroke Committee / SRCA Medical Director

## 1. Purpose

To ensure timely, safe, and appropriate transport of patients with suspected acute stroke to the most appropriate stroke center while minimizing delays to reperfusion therapy.

## 2. Scope

This protocol applies to:

- Saudi Red Crescent Authority (SRCA)
- Emergency Department staff
- Stroke Team
- Transfer Center
- Critical Care Transport Teams
- Air Medical Transport (when applicable)

## 3. Transport Eligibility

Patients shall be transported under the Stroke Alert pathway if they have:

- Sudden onset of focal neurological deficit
- Positive validated stroke screening tool (FAST, BE-FAST, CPSS, etc.)
- Last Known Well (LKW) documented or estimated
- Symptoms suggestive of acute ischemic or hemorrhagic stroke

## 4. Destination Selection

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### A. Primary Stroke Center (PSC)

Transport to the nearest PSC when:

- Suspected acute stroke without evidence of large vessel occlusion (LVO)
- Estimated transport time to Comprehensive Stroke Center (CSC) would cause significant treatment delay
- Patient is a potential candidate for intravenous thrombolysis

### B. Comprehensive Stroke Center (CSC)

Transport directly to a CSC when:

- Positive LVO screening (FAST-ED, RACE, LAMS, VAN, etc.)
- Suspected need for mechanical thrombectomy
- Severe stroke (e.g., NIHSS  $\geq 6$ , according to regional protocol)
- Intracerebral hemorrhage requiring neurosurgical care
- Regional SRCA destination criteria are met

## 5. Transport Priorities

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SRCA should:

- Minimize scene time (target  $\leq 15$  minutes)
- Obtain IV access if it does not delay transport
- Check blood glucose
- Monitor airway, breathing, circulation, and neurological status
- Avoid unnecessary procedures that delay arrival

## 6. Hospital Pre-Notification

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SRCA shall notify the receiving hospital before arrival and provide:

- Estimated time of arrival (ETA)
- Patient age and sex
- Last Known Well
- Stroke screening result
- LVO score (if performed)
- Blood glucose
- Blood pressure
- Current neurological deficits
- Anticoagulant use (if known)
- Airway status

## 7. Interfacility Transfer Protocol

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Patients requiring higher-level stroke care shall be transferred without unnecessary delay.

### Indications Include

- Mechanical thrombectomy
- Neurosurgical intervention
- Neurocritical care
- Advanced neuroimaging unavailable at referring hospital
- Complex stroke management beyond local capability

### Prior to Transfer, the Referring Hospital Shall

- Stabilize airway, breathing, and circulation
- Perform CT brain (and CTA if available)
- Send imaging electronically whenever possible
- Complete transfer documentation
- Notify the receiving stroke team
- Arrange the fastest appropriate transport

### Role of the Resource Control Center (RCC)

For peripheral hospitals without on-site stroke services, the Resource Control Center (RCC) is the designated arrangement center for coordinating the interfacility transfer of acute stroke patients.

- The referring hospital's Emergency Department (ED) physician contacts the RCC directly for any suspected or confirmed acute stroke patient at a facility without stroke services.
- The RCC connects the referring ED physician with the RCC-Neurology on-call physician for case discussion and clinical/imaging review.
- Transfer shall not be arranged until the patient has been formally accepted by the RCC-Neurology on-call physician.
- Once acceptance is obtained, the RCC coordinates receiving-center assignment, bed availability, and transport arrangements.
- The referring hospital completes pre-transfer stabilization and documentation (see "Prior to Transfer, the Referring Hospital Shall" above) in parallel with RCC coordination.

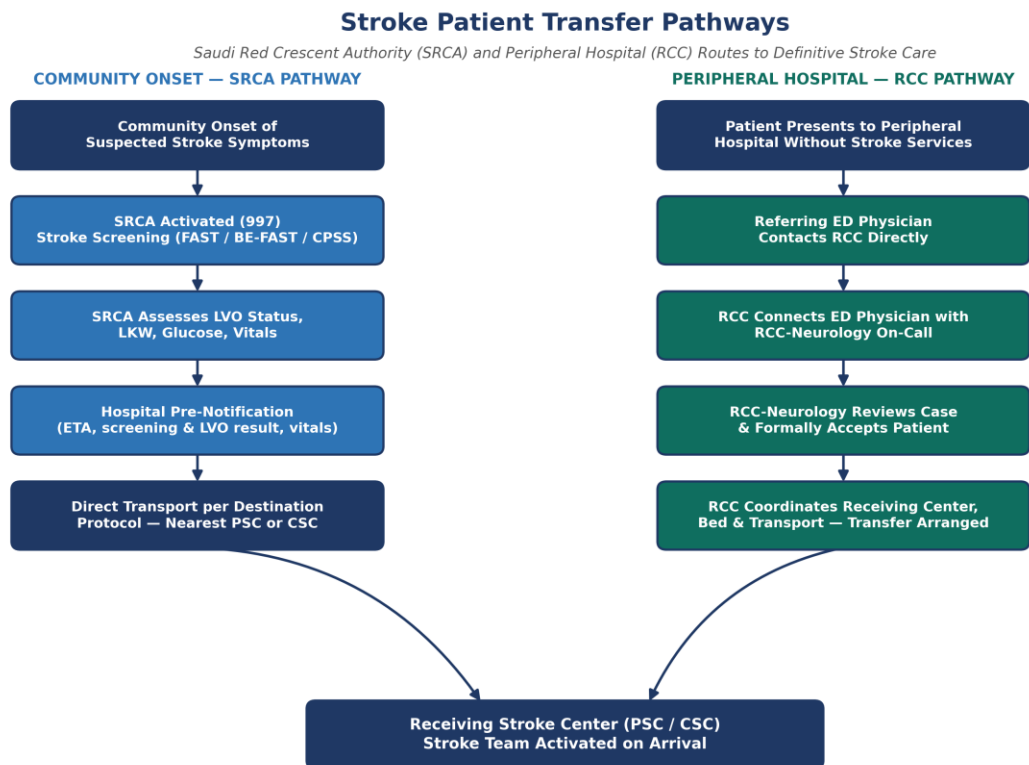


Figure 1. Stroke patient transfer pathways — community onset via SRCA (left) and peripheral hospital referral coordinated through the RCC (right), both converging on the receiving stroke center.

## 8. Mode of Transport

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### Ground Ambulance

Preferred when transport time is acceptable and rapid access is available.

### Air Ambulance

Consider when:

- Long transport distance
- Time-sensitive thrombectomy candidate
- Remote location
- Ground transport would significantly delay definitive treatment

## 9. Monitoring During Transport

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Monitor and document:

- Blood pressure
- Heart rate
- Respiratory rate
- Oxygen saturation
- Glasgow Coma Scale (GCS)
- Neurological status
- Blood glucose (if indicated)

### Maintain

- SpO<sub>2</sub> ≥ 94%
- Normoglycemia
- Airway protection
- Intravenous access

## 10. Documentation

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Record:

- Symptom onset / Last Known Well
- Stroke screening tool results
- LVO assessment
- Vital signs
- Blood glucose
- Treatments provided
- Time of SRCA activation
- Departure time
- Arrival time
- Hospital prenotification time
- Any clinical deterioration during transport

## 11. Quality Indicators

| Indicator                                   | Target                   |
|---------------------------------------------|--------------------------|
| SRCA prenotification                        | $\geq 95\%$              |
| Last Known Well documented                  | $\geq 95\%$              |
| Blood glucose documented                    | $\geq 95\%$              |
| Scene time                                  | $\leq 15$ minutes        |
| Appropriate destination compliance          | $\geq 90\%$              |
| Door-In–Door-Out (transfer hospitals)       | $\leq 60$ minutes (goal) |
| Transfer imaging available before departure | $\geq 95\%$              |

## 12. Staff Education

All SRCA and transport personnel shall receive annual education on:

- Stroke recognition
- Stroke screening tools
- LVO assessment
- Destination decision-making
- Safe transport practices
- Hospital prenotification
- Documentation requirements

*Competency should be assessed annually.*

## References

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- American Heart Association / American Stroke Association. Guidelines for the Early Management of Patients With Acute Ischemic Stroke.
- Saudi Red Crescent Authority (SRCA). National Ambulance and Prehospital Stroke Response Standards.
- Saudi Red Crescent Authority (SRCA) Stroke Destination Policy.
- Hospital Stroke Program Policies.

## SRCA & AHA/ASA Regulatory Alignment and Accreditation

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### Regulatory Concordance & Accreditation Support

- **Guideline concordance:** Destination selection, LVO screening, and hospital pre-notification criteria in Sections 3–6 are drawn directly from the 2026 AHA/ASA Guideline for the Early Management of Patients with Acute Ischemic Stroke.
- **SRCA regulatory alignment:** Transport eligibility, scene-time targets, destination selection, and hospital pre-notification requirements in Sections 3–6 are aligned with Saudi Red Crescent Authority (SRCA) national ambulance response and prehospital stroke transport standards, ensuring dual compliance with national regulatory requirements and international clinical guidelines.
- **Accreditation support:** Prenotification, LKW, and Door-In–Door-Out targets tracked in Section 11 provide the auditable performance data required for Comprehensive Stroke Center (CSC) certification (American Stroke Association International Stroke Center Certification Advanced Certification) and AHA Get With The Guidelines®-Stroke reporting.
- **Reference standard:** American Heart Association / American Stroke Association Guidelines for the Early Management of Patients With Acute Ischemic Stroke; Saudi Red Crescent Authority (SRCA) National Ambulance and Prehospital Stroke Response Standards (see References).

Approval & Sign-Off

| Role                                        | Name & Title | Signature | Date |
|---------------------------------------------|--------------|-----------|------|
| Prepared By                                 |              |           |      |
| Reviewed By<br>(Stroke Program Coordinator) |              |           |      |
| Approved By<br>(SRCA Medical Director)      |              |           |      |
| Approved By<br>(Stroke Committee Chair)     |              |           |      |

Effective date: \_\_\_\_\_ Next review date: \_\_\_\_\_